Arya Mohammadi

+1(424)-204-2995 | aryamohammadi.dev@gmail.com | linkedin.com/in/aryamshahi | github.com/aryamohammadi | aryamohammadi.github.io

Education

University of California, San Diego — Expected Graduation – **Jun. 2027**
B.S. Mathematics & Computer Science — *GPA: 3.7/4*

Relevant Coursework: Advanced Data Structures, Design & Analysis of Algorithms, Software Engineering, Systems Programming, Artificial Intelligence, Data Science Principles, Graph Theory, Probability & Statistics

Technical Skills

Languages: Python, C++, C, Rust, JavaScript, TypeScript, SQL, Java
Frameworks & Tools: React, Next.js, Node.js, Express, FastAPI, OpenSearch, MongoDB, Docker, Git, Linux, Jest, GoogleTest
Concepts: REST APIs, CI/CD, Unit & E2E Testing, Authentication, RAG, Hybrid Retrieval, Model Context Protocol (MCP)

Experience

IBM — Jun. 2026 – Sept. 2026
Software Developer Intern — *Austin, TX*

- Improved retrieval performance in a production RAG pipeline (Python, OpenSearch) by implementing score normalization and safely deploying vector search behind feature flags, increasing Recall@3 by 55%.
- Developed an automated retrieval evaluation pipeline to benchmark 1,000+ labeled queries using NDCG@10 and MRR, integrating regression checks into CI to reduce manual ranking validation effort.
- Built a RAG-powered onboarding agent using Python, watsonx Orchestrate ADK, and IBM Granite to retrieve internal documentation and guide new hires through setup workflows; adopted by the Orchestrate team, reducing onboarding support requests by 45%.
- Developed a reusable Python/pytest API testing framework to automate regression testing across 80+ watsonx Orchestrate REST endpoints, leveraging parameterized tests and shared fixtures to validate authentication, error handling, and response schemas; integrated into existing CI/CD pipelines for team-wide use.
- Placed 3rd out of 184 teams at IBM's internal hackathon by building a semantic cache (FastAPI, MCP) that reused responses from an internal LLM coding assistant, achieving a 40% cache hit rate and reducing response latency by 37%.

Arya Electric LLC | **aryaelectric.com** | **engineering write-up** — Jun. 2025 – Present
Software Engineer & Co-Founder — *Irvine, CA*

- Architected and deployed a full-stack application using Next.js, TypeScript, PostgreSQL, and Twilio, integrating portfolios, automated quote requests, & event-driven SMS/email notifications; facilitated $30K+ in booked jobs and increased inbound leads by 25%.
- Built an automated quote intake system using Next.js, TypeScript, and Claude Haiku to extract project scope, budgets, and timelines from customer messages; used Zod to validate LLM outputs, reducing intake processing time by 40%.
- Built and deployed a Next.js and Firebase admin dashboard with secure authentication, image compression, and automated photo management, reducing publishing time by 80% and eliminating manual redeployments.

University of California, Riverside — Sept. 2024 – Jun. 2025
Computer Science and Mathematics Tutor — *Riverside, CA*

- Guided **400+** students over **600** hours through data structures and algorithms **(C++)**, improving student understanding of memory management, recursion, and algorithmic complexity.
- Debugged **100+** codebases, diagnosing segmentation faults, memory leaks, & pointer errors; authored conceptual practice problems.

Projects

Brickt — Co-founder (Aug. 2026 – Present) | *TypeScript, Next.js, React, PostgreSQL* | GitHub

- Co-founded Brickt, a construction management web app; designed, architected, and built the application from concept to staging deployment in 7 weeks.
- Designed and implemented centralized role-based access control (RBAC) using Node.js, Express, and PostgreSQL, enforcing server-side permissions and subcontractor-level bid isolation; developed automated tests covering all defined authorization scenarios.
- Shipped production software as a founding engineer on a 3-person team, leveraging AI coding agents to accelerate development while enforcing code quality through code reviews, coding standards, and automated CI checks.

AgentSight — Cursor IDE Support (Open Source) | *Rust, SQLite, Python* | PR #149 (merged) | GitHub

- Engineered a read-only SQLite tracing backend for AgentSight, bypassing eBPF compatibility limitations and leveraging indexed key-range queries to recover AI request traces, including model identifiers, latency, and token usage, from Cursor's 500 MB database.
- Added Cursor session discovery to `agent-session`, a published Rust library, without breaking its public API: existing sessions appear with zero configuration, while delegated sub-agent runs neither create phantom sessions nor leave the session cache stale.
- Co-authored PR #149, adding 15 automated tests and resolving a Rust crate packaging failure in CI; merged and released in v1.0.4.